Essential VS Code Extensions (12 Must-Haves)

Core Development (Install First):

1. **GitHub Copilot** – AI pair programmer
2. **Prettier** – Code formatter
3. **Live Share** – Real-time collaboration
4. **Error Lens** – Inline error highlighting

HTML/CSS Productivity:

5. **Auto Rename Tag** – Auto-rename HTML/XML tags
6. **Auto Close Tag** – Automatically add closing tags
7. **CSS Peek** – Jump to CSS definitions from HTML

JavaScript/TypeScript:

8. **ESLint** – JavaScript/TypeScript linter
9. **JavaScript (ES6) Code Snippets** – Smart code templates

Git & Version Control:

10. **GitLens** – Git supercharger
11. **Git Graph** – Visual commit history browser

Visual & Productivity:

12. **Material Icon Theme** – File/folder icons
13. **Bracket Pair Colorizer** – Color-coded brackets
14. **TODO Highlight** – Highlight TODO comments

Bonus: 3 Performance Extensions

15. **Import Cost** – Show size of npm packages
16. **Code Spell Checker** – Spell checking for code
17. **Thunder Client** – API testing inside VS Code

Quick Installation Steps:

1. Open VS Code
2. Click Extensions icon (sidebar, or Ctrl+Shift+X)
3. Search each extension name
4. Click "Install"
5. Restart VS Code after installing all

Git & GitHub: Essential Concepts

Version Control Explained

What is Version Control?

Think of it as "Google Docs for code" - it tracks every change, lets multiple people work together, and saves every version so you can go back if you mess up.

Why You Need It:

1. **Backup:** Never lose work
2. **Collaboration:** Work with others without breaking things
3. **Experimentation:** Try new ideas safely
4. **Portfolio:** GitHub = your developer resume

Key Terms:

- **Repository (Repo):** Your project folder that Git tracks
- **Commit:** Save point (like saving a game)
- **Push:** Upload to GitHub
- **Pull:** Download from GitHub
- **Branch:** Parallel universe for trying new features
- **Merge:** Combine changes from different branches

10 Essential Git Commands

Daily Workflow (Use These 95% of the Time)

1. git status – Shows what changed since last commit.
2. git add . – Stages all changes for commit.
3. git commit -m "message" – Saves changes with description.
4. git push – Uploads commits to GitHub.
5. git pull – Downloads updates from GitHub.

Setup & Fixes

6. git clone <url> – Copies a repository from GitHub to your computer.
7. git log --oneline – Shows commit history in compact view.
8. git checkout -- <file> – Discards changes in a specific file.
9. git branch – Lists all branches (asterisk shows current branch).
10. git merge <branch> – Combines another branch into current branch.